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MATHEMATICS

STANDARD FIVE

Monthly test - September 2024
Answer Key and Worked Solutions

Subject	Mathematics
Code	Not printed in source
Level	Standard Five
Year	2024
Exam Board / Issuer	SPES
Exam	Monthly test - September
Type	Answer key and worked solutions
Questions	33 numbered questions/subparts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S5-2024-SPES-September-Monthly-test.pdf

Notes about the source paper

Items with missing information or printing errors are explained in their solutions. A model response is an example, rather than a prescribed official answer.

Calculations

1(i). Add $2138 + 1471$.

Step 1: 1 place: $8 + 1 = 9$; write 9.

Step 2: 10 place: $3 + 7 = 10$; write 0, carry 1.

Step 3: 100 place: $1 + 4 + \text{carry } 1 = 6$; write 6.

Step 4: 1000 place: $2 + 1 = 3$; write 3.

Step 5: $2138 + 1471 = 3609$.

Final Answer: 3609

1(ii). Subtract $367148 - 243128$.

Step 1: 1 place: $8 - 8 = 0$.

Step 2: 10 place: $4 - 2 = 2$.

Step 3: 100 place: $1 - 1 = 0$.

Step 4: 1000 place: $7 - 3 = 4$.

Step 5: 10000 place: $6 - 4 = 2$.

Step 6: 100000 place: $3 - 2 = 1$.

Step 7: $367148 - 243128 = 124020$; check $124020 + 243128 = 367148$.

Final Answer: 124020

1(iii). Multiply 989×53 .

Step 1: Split into place values: $900 \times 53 = 47700$; $80 \times 53 = 4240$; $9 \times 53 = 477$.

Step 2: Add the partial products: 52417.

Final Answer: 52417

1(iv). Divide 4984 by 56.

Step 1: $498 \div 56$: write 8; $8 \times 56 = 448$; $498 - 448 = 50$.

Step 2: $504 \div 56$: write 9; $9 \times 56 = 504$; $504 - 504 = 0$.

Step 3: $56 \times 89 = 4984$.

Step 4: $4984 \div 56 = 89$.

Final Answer: 89

4(i). Population 1450000, women 803000. Men?

Step 1: 1 place: $0 - 0 = 0$.

Step 2: 10 place: $0 - 0 = 0$.

Step 3: 100 place: $0 - 0 = 0$.

Step 4: 1000 place: regroup 1 from the next place; $10 - 3 = 7$.

Step 5: 10000 place: $4 - 0 = 4$.

Step 6: 100000 place: regroup 1 from the next place; $14 - 8 = 6$.

Step 7: 1000000 place: $0 - 0 = 0$.

Step 8: $1450000 - 803000 = 647000$; check $647000 + 803000 = 1450000$.

Final Answer: 647000 men

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Mathematics

1(ix). Subtract 14.13 - 8.16.

Step 1: Borrow in hundredths: $13 - 6 = 7$; then 0 tenths must borrow 1 whole, $10 - 1 = 9$.

Step 2: Whole $13 - 8 = 5$.

Final Answer: 5.97

1(v). Add $\frac{2}{3} + \frac{4}{5}$.

Step 1: Common denominator 15: $\frac{10}{15} + \frac{12}{15}$.

Step 2: $10 + 12 = 22$; $\frac{22}{15} = 1 \frac{7}{15}$.

Final Answer: $1 \frac{7}{15}$

1(vi). Subtract $2 \frac{5}{9} - \frac{1}{2}$.

Step 1: $2 \frac{5}{9} = \frac{23}{9} = \frac{46}{18}$; $\frac{1}{2} = \frac{9}{18}$.

Step 2: $46 - 9 = 37$; $\frac{37}{18} = 2 \frac{1}{18}$.

Final Answer: $2 \frac{1}{18}$

1(vii). Multiply $\frac{3}{4} \times 1 \frac{3}{5}$.

Step 1: $1 \frac{3}{5} = \frac{8}{5}$.

Step 2: $3 \times 8 = 24$; $4 \times 5 = 20$.

Step 3: $\frac{24}{20} = \frac{6}{5} = 1 \frac{1}{5}$.

Final Answer: $1 \frac{1}{5}$

1(viii). Divide $\frac{3}{4}$ by 3.

Step 1: Multiply by reciprocal $\frac{1}{3}$.

Step 2: $(\frac{3}{4}) \times (\frac{1}{3}) = \frac{3}{12} = \frac{1}{4}$.

Final Answer: $\frac{1}{4}$

1(x). Multiply 3.05×0.3 .

Step 1: $305 \times 3 = 915$.

Step 2: Total 3 decimal places.

Final Answer: 0.915

2(i). Continue 1,4,9,16,[].

Step 1: Squares $1^2, 2^2, 3^2, 4^2$; next 5^2 .

Step 2: $5 \times 5 = 25$.

Final Answer: 25

2(ii). Read CDXLVII in words.

Step 1: $CD=400$; $XL=40$; $VII=7$.

Step 2: $400 + 40 + 7 = 447$.

Final Answer: Four hundred and forty-seven

2(iii). Value of 9 in 392527?

Step 1: 9 in ten-thousands.

Step 2: $9 \times 10000 = 90000$.

Final Answer: 90000

2(iv). Write eight hundred thousand five hundred and fifty-seven in figures.

Step 1: $800000 + 500 + 50 + 7 = 800557$.

Final Answer: 800557

2(v). Subtract square root 64 from square root 225.

Step 1: $8^2=64$; $15^2=225$.

Step 2: $15 - 8 = 7$.

Final Answer: 7

2(vi). Find the LCM of 6,12,24.

Step 1: $6=2 \times 3$; $12=2^2 \times 3$; $24=2^3 \times 3$.

Step 2: Highest powers $2^3 \times 3=24$.

Final Answer: 24

3(i). Name place of underlined second 1 in 1143578.

Step 1: From left: millions 1, hundred-thousands 1, ten-thousands 4,...

Step 2: The underline is below the second 1.

Final Answer: Hundred-thousands

3(ii). Write 50000000 +5000000 +500000 +0 +1000 +100 +0 +6 in short form.

Step 1: $50000000 + 5000000 + 500000 + 1000 + 100 + 6 = 55501106$.

Final Answer: 55501106

3(iii). Which prime equals an even number so their difference is 0?

Step 1: The only even prime is 2.

Step 2: $2 - 2 = 0$.

Final Answer: 2

4(ii). Brick length 24 cm 60 mm; log 3 cm 70 mm. Total length?

Step 1: $60 \text{ mm} = 6 \text{ cm}$; $70 \text{ mm} = 7 \text{ cm}$.

Step 2: Brick $24 + 6 = 30 \text{ cm}$; log $3 + 7 = 10 \text{ cm}$.

Step 3: $30 + 10 = 40$.

Final Answer: 40 cm (400 mm)

4(iii). How many even numbers between 73 and 85?

Step 1: 74,76,78,80,82,84.

Final Answer: 6

5(i). Find the GCF of 14,36,42.

Step 1: $14 = 2 \times 7$; $36 = 2^2 \times 3^2$; $42 = 2 \times 3 \times 7$.

Step 2: Only common prime is 2.

Final Answer: 2

5(ii). Runner starts 8:55 a.m.and runs 1 hour 40 minutes. Finish time?

Step 1: $8:55 + 1 \text{ hour} = 9:55$.

Step 2: 40 minutes more reaches 10:35.

Final Answer: 10:35 a.m.

5(iii). Days in 3 leap years?

Step 1: Each has 366 days.

Step 2: $3 \times 366 = 1098$.

Final Answer: 1098 days

6(i). Buy 15 books at 600 sh, 8 pens at 500 sh, 5 envelopes at 200 sh, 3 erasers at 200 sh. Total?

Step 1: $15 \times 600 = 9000$.

Step 2: $8 \times 500 = 4000$.

Step 3: $5 \times 200 = 1000$.

Step 4: $3 \times 200 = 600$.

Step 5: $9000 + 4000 + 1000 + 600 = 14600$.

Final Answer: Sh 14600

6(ii). Number of primes between 30 and 52?

Step 1: Primes 31, 37, 41, 43, 47.

Step 2: All other candidates have a divisor 2, 3, 5 or 7.

Final Answer: 5

6(iii). Find square of 491.

Step 1: $491 \times 491 = 241081$.

Final Answer: 241081

7(i). Triangle base 4.5 m, perpendicular height 4 m. Area?

Step 1: $4.5 \times 4 = 18$.

Step 2: $18 / 2 = 9$.

Final Answer: 9 m²

7(ii). Trapezium parallel sides 7 cm,10 cm; height 4 cm. Area?

Step 1: $\text{Area} = (\text{parallel sides sum}) \times \text{height} / 2$.

Step 2: $7 + 10 = 17$.

Step 3: $17 \times 4 / 2 = 34$.

Final Answer: 34 cm²

7(iii). Parallelogram top 600 m, bottom 300(no unit), height 300 m. Find area.

Step 1: Opposite bases must be equal, so 600 and 300 conflict.

Step 2: If 600 m is intended base: $600 \times 300 = 180000 \text{ m}^2$.

Step 3: If 300 m is intended base: $300 \times 300 = 90000 \text{ m}^2$.

Final Answer: Conditional 180000 m² using top 600 m, or 90000 m² using bottom 300 m; no consistent unique area.

8(i). Find perimeter of square side 10.5 cm.

Step 1: $4 \times 10.5 = 42$.

Final Answer: 42 cm

8(iii). How many symmetry lines in the pictured square?

Step 1: Vertical, horizontal, and 2 diagonals.

Final Answer: 4

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Source Checks

Darasa Huru - SPES Standard Five 5 Monthly Exams September 2024